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Early Warning of Suspected Doping from Biological Passport Based on Multivariate Trends

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ABSTRACT

The indirect identification of doping in sports can be performed by assessing athletes' hematological perturbations from the analysis of blood collected on different occasions. Because prosecution for doping based on this information requires expensive and time-consuming interpretation of blood analysis results by various expert hematologists, mathematical data screening is performed to decide which cases should be forwarded to hematologists. The current Bayesian and univariate screening of data does not process the multivariate trends of blood parameters or take the time interval between samplings into account. This work presents a computational tool that overcomes these limitations by calculating a single score, the hematological perturbation index (HP_{Ix}), for which a threshold is defined above which hematologists should be asked to assess the athlete's biological passport. The doping detection from this index, normalized for days difference between samplings (HP_{Ix}^{N}) based on 3, 4 or 5 consecutive samplings, is associated with true positive result rates (TP) not below 98 % and false positive result rates (FP) less than 0.9 %. Therefore, this tool can be useful as an early warning system of hematological perturbations to decide which athletes should be more closely monitored and which biological passports should be forwarded to hematologists for medical interpretation of data.

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Introduction

The direct detection of doping substance use through the analysis of urine samples is possible only if the control is performed close to drug administration. Therefore, although direct doping control produces unchallengeable evidence of fraud, only when doping is continued or frequent (i.e., not occasional) is there a non-negligible probability of fraud detection [1–4].

The athlete biological passport (ABP) represents a new paradigm of fighting doping in sports based not on the direct detection of doping substances or methods, but on detecting variations of key doping biomarkers through the longitudinal/temporal monitoring of such markers. In the hematological module (steroidal and endocrine modules will not be addressed here), this implies monitoring the variation of several hematological parameters [1, 2].

However, some confounding factors like disease, altitude, or recent exercise can also produce relevant changes to the hematological parameters [5–9]. Therefore, the interpretation of hematological parameter variations is performed by medical doctors who consider all the available information required to understand if any observed abnormality in the profiles can be attributed to doping. In fact, the prosecution of an athlete for doping based on his or her biological passport must be supported by convergent reports from various independent expert hematologists [10–13].

In the current version of the ABP, the biomarkers are automatically processed using an Adaptive Model that runs on software developed by the World Anti-Doping Agency (WADA). The Adaptive Model predicts, from the data of the tested individual, an expected range within which the marker values should fall assuming a normal physiological condition. Outliers correspond to those values outside a 99 % range limited by the defined 0.5th and 99.5th percentiles associated with a 1:100 chance or less that the result is due to normal physiological variation. This Adaptive Model takes the information of each blood parameter separately considering a Bayesian evaluation of prior information updated by each new measurement result of athletes' blood. The mathematical detection of relevant hematological deviations requires the collection of information on several occasions extended over a long enough period to guarantee that the same physiological status is not monitored [1, 13].

The univariate and Bayesian management of information is disadvantageous because it does not allow the easy assessment of hematological multivariate trends and it requires a large number of monitoring occasions before decisions on the biological passport can be taken. The hematologists take into account the correlation of data in their interpretation of the biological passport.

Alladio et al. [14] assessed the multivariate difference of endogenous anabolic androgenic steroid (EAAS) levels in the urine of individuals who administered or did not administer EAAS. The two groups were studied using principal component analysis and Hotelling's T^2 tests [15–18]. The differentiation of the groups was not based on the variation of these parameters over time.

The current work suggests a fully automatic multivariate assessment of biological passport perturbations through the determination of a single score designated the hematological perturbation index (HP_{Ix}). The comparison of HP_{Ix} with a threshold can trigger the consultation of expert hematologists at a low rate of false suspicions. This multivariate assessment can be performed with results of just three monitoring occasions, and the quality of the tool's output increases when four or five sequential samplings are considered. When the tool developed is applied to more data than required for detecting physiological perturbations, it identifies the period in which perturbations were relevant or larger.

The developed methodology was successfully used to distinguish athletes' biological passports with large or negligible hematological perturbations confirmed by hematologists.

A computation tool was also developed to estimate the true positive (TP) and false positive (FP) result rate of the screening by the Monte Carlo simulation of hematological patterns equivalent to the ones studied by taking into account the uncertainty of parameters determination. The TP and FP were combined in the likelihood ratio of identifications (LR = TP/FP) that quantifies the identification uncertainty [19–21].

Theory

Multivariate information of the biological passport

The detection of doping cases from the biological passport, i.e., from hematological data obtained on different occasions, involves assessing blood parameter variations. Ideally, this assessment should take the temporal difference between collected data into account. The current Bayesian assessment of the biological passport does consider the time interval between blood collections.

Because doping induces the correlated variation of various blood parameters, this correlation should also be considered in the interpretation of data. The direct or inverse correlation of an adequate pair of parameter values can be extremely informative as to the relevance of the physiological processes involved. In fact, hematologists examine the biological passport by looking at the synchronism of some trends.

This work proposes an automatic evaluation of the correlation between blood parameters useful to distinguish negligible and meaningful trends; meaningful trends are attributable to relevant physiological perturbations. This strategy was applied to the assessment of hematological data from 3, 4, or 5 sampling occasions.

Multivariate trends of three samplings

For simplicity, the assessment of three consecutive sampling occasions is detailed; the following steps are completed:

Step 1

Hematological data of N individuals are collected, where for each individual q (q = 1 to N) data from m_q sampling occasions are acquired. The number of blood samplings can vary with the individual but m_q cannot be smaller than three ($m_q \geq 3$) to allow assessing trends of three consecutive samplings. The sampling index j of individual q represents the jth sampling occasion of blood from that person and varies between 1 and m_q. The data from individual q and blood sampling occasion j is used to build a vector that combines the results of respective n blood parameters, b: (b_{qj1},..., $b_{qji,}$ ..., $b_{qjn,}$), where i is the index of the ith blood parameter (i = 1 to n). The vector (b_{qj1},..., $b_{qji,}$ ..., $b_{qjn,}$) can be represented by the symbol $\mathbf{b_{qj}}$; vectors are identified in bold. The hematological module of the biological passport includes the determination of eight blood parameters (n = 8): hemoglobin concentration (HGB) (g/L), hematocrit volume percentage (HCT) (%), red blood cell counts (RBC) (× 10^9 units/L), reticulocyte counts (RET) (units/nL), reticulocyte volume percentage (RET %) (%), mean corpuscular volume (MCV) (fL), mean corpuscular hemoglobin (MCH) (pg), and mean corpuscular hemoglobin concentration (MCHC) (g/dL).

Step 2

The result of each blood parameter (b_{qji}) is normalized considering the mean ($\overline{b}_i$) and standard deviation (s_{bi}) of b_i values observed in various individuals of a population equivalent to the assessed athletes. New vectors of the normalized parameters are produced ($\hat{b}_{qj1}$,...,$\hat{b}_{qji}$,...,$\hat{b}_{qjn}$) where $\hat{b}_{qji}$ is the normalized value of blood parameter b_i of individual q on occasion j: $\hat{b}_{qji} = (b_{qji} - \overline{b}_i)/s_{bi}$. The vector ($\hat{b}_{qj1}$,...,$\hat{b}_{qji}$,...,$\hat{b}_{qjn}$) can be represented by the symbol $\hat{\mathbf{b}}_{\mathbf{qj}}$.

Athletes randomly selected for doping control who do not represent a specific health condition or doping situation can yield an adequate reference population for blood parameter normalization. Otherwise, this normalization can be based on the data of non-athletes ideally from an age range and health condition equivalent to that of athletes. ▶ **Table 1** presents the $\overline{b}_i$ and s_{bi} of the reference data used. This normalization was performed although three blood parameters, namely RET, RET % and MCHC, are not normally distrib-

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▶ **Table 1** Mean value and standard deviation of studied blood parameters used in the normalization of the parameters. These values were estimated from 1082 results of the analysis of each parameter from blood samples of 127 athletes.

Parameter	$\bar{b}_i$	S_{bi}	Parameter	$\bar{b}_i$	S_{bi}
HGB (g/L)	145	10	RET % (%)	1.12	0.43
HCT (%)	42.2	2.8	MCV (fL)	89.8	4.2
RBC ($\times 10^9$ units/L)	4.71	0.41	MCH (pg)	30.8	1.4
RET (units/nL)	53	20	MCHC (g/dL)	34.34	0.90

$\bar{b}_i$, mean value; s_{bi}, standard deviation.

uted (p-value < 0.01; Kolmogorov Smirnov test). However, these three parameters are log-normally distributed (p-value > 0.01) with a deviation from normality that does not affect the general purpose of the normalization, i.e., to eliminate the scale effect on data treatment.

Step 3

The Euclidean distances of normalized vectors of various sampling occasions of each individual are estimated. Eq.▶ **(1)** represents the calculation of the Euclidean distance, $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})$, between vectors $\hat{\mathbf{b}}_{qj}$ and $\hat{\mathbf{b}}_{q(j+1)}$ of two consecutive samplings (j^{th} and $(j+1)^{th}$ sampling occasions) from individual q.

$$d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) = \sqrt{\begin{array}{l}(\hat{b}_{qj1} - \hat{b}_{q(j+1)1})^2 + (\ldots) \\ + (\hat{b}_{qji} - \hat{b}_{q(j+1)i})^2 + (\ldots) \\ + (\hat{b}_{qjn} - \hat{b}_{q(j+1)i})^2\end{array}} \quad (1)$$

Step 4

For each three consecutive blood collection occasions of individual q (sampling occasions j, j + 1 and j + 2), the Euclidean distances are estimated between the jth and $(j+1)^{th}$, $(j+1)^{th}$ and $(j+2)^{th}$, and j^{th} and $(j+2)^{th}$ vectors, i.e., $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})$, $d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)})$, and $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)})$, respectively. The $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)})$ is calculated to assess if the third vector moves away from or closest to the first vector. These three distances are summed to quantify the multivariate variation of blood parameters for the studied group of samplings, $\Sigma(j \rightarrow 2)$ (Eq.▶ **(2)**).

$$\Sigma(j \rightarrow 2) = d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) + d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)}) \quad (2)$$

The $\Sigma(j \rightarrow 2)$ increases with the size of multivariate deviations in the biological passport.

For instance, for an individual for whom data are available from 6 sampling occasions, 4 $\Sigma(j \rightarrow 2)$ values are determined between the 1st and 3rd, 2nd and 4th, 3rd and 5th, and 4th and 6th sampling occasions.

Step 5

For each individual, the largest $\Sigma(j \rightarrow 2)$ is estimated, identified as $\Sigma_{Max}(j \rightarrow 2)$, which represents the HP_{Ix} of the athlete's hematological system.

Step 6

From the population of studied athletes, hematologists identify individuals with perturbed and unperturbed hematological systems.

Step 7

Given the uncertainty of the blood parameters determination, hematological histories are simulated equivalent to the ones available using the Monte Carlo method. Other possible blood parameter values are simulated by taking the measured value b_{qji}, the standard deviation of measurement reproducibility (s_{Ri}), and simulated one-tailed t values (t_s) of the Student's distribution for the degrees of freedom of (s_{Ri}) and a random probability level between 0 and 1 (U(0,1)): $b_{qji} + s_{Ri}\, t_s$.

Step 8

For each new hematological history simulation, $\Sigma_{Max}(j \rightarrow 2)$ is determined. From simulated $\Sigma_{Max}(j \rightarrow 2)$ of athletes with confirmed abnormal perturbations of their hematological system, a threshold was defined, $\Sigma_{TH}(j \rightarrow 2)$, positioned below more than 98 % of the simulated $\Sigma_{Max}(j \rightarrow 2)$. The fraction of simulated $\Sigma_{Max}(j \rightarrow 2)$ values above $\Sigma_{TH}(j \rightarrow 2)$ estimates the true positive result rate (TP) of perturbation detection.

Step 9

After $\Sigma_{TH}(j \rightarrow 2)$ is defined, the number of simulated $\Sigma_{Max}(j \rightarrow 2)$ above the threshold from athletes with negligible hematological perturbations is determined. The fraction of these values, given the total number of simulations, quantifies the false positive result rate (FP) of identifications based on the defined threshold.

Step 10

Because this tool is used to automatically screen the biological passports to select which profiles should be forwarded to hematologists or which athletes should be monitored at the next opportunity, the tool is adequate if TP is larger than 98 % and FP not larger than 2 %.

The estimated TP and FP of perturbation identification can be combined in the likelihood ratio (LR = TP/FP), but no target value is defined for the LR. If TP is 98 % and FP is 2 %, the LR is 49, but larger LRs are possible if TP is lower that 98 % and FP also reduces. A likelihood ratio of 49 specifies that the identification is 49 times more likely true than false. This metric is frequently reported as the identification uncertainty [19–21].

Multivariate trends of four or five samplings

The described strategy for assessing multivariate trends from three consecutive blood samplings was applied to evaluations from four or five consecutive samplings after changing some data processing details. The following changes in data processing stages were performed.

Step 1

For studying multivariate trends of 4 or 5 consecutive blood samplings, $m_q \geq 4$ and $m_q \geq 5$, respectively.

Step 2

This stage was not changed.

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▶ **Table 2** Relative reproducibility standard deviations (s'_R) of blood parameters determination.

Parameter	s'_R(%)	Parameter	s'_R(%)
Hemoglobin concentration (HGB)	0.567	Reticulocyte volume percentage (RET%)	10.0
Hematocrit volume percentage (HCT)	1.25	Mean corpuscular volume (MCV)	1.33
Red blood cell counts (RBC)	1.12	Mean corpuscular hemoglobin (MCH)	1.30
Reticulocyte counts (RET)	10.0	Mean corpuscular hemoglobin concentration (MCHC)	1.33

Stages 3 and 4

For four samplings the following Euclidean distances were calculated: $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})$, $d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)})$, $d(\hat{\mathbf{b}}_{q(j+2)}, \hat{\mathbf{b}}_{q(j+3)})$, $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)})$, and $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+3)})$. In this case, the sum of Euclidean distances $\Sigma(j\rightarrow3)$ is estimated by Eq.▶ **(3)**.

$$\Sigma(j\rightarrow3) = d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) + d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)}) + d(\hat{\mathbf{b}}_{q(j+2)}, \hat{\mathbf{b}}_{q(j+3)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+3)}) \quad (3)$$

For five samplings, the same Euclidean distances were estimated for four samplings plus $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+4)})$, $d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+4)})$, and $d(\hat{\mathbf{b}}_{q(j+3)}, \hat{\mathbf{b}}_{q(j+4)})$. In this case, the sum of Euclidean distances, $\Sigma(j\rightarrow4)$, is estimated by Eq.▶ **(4)**.

$$\Sigma(j\rightarrow4) = d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) + d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)}) + d(\hat{\mathbf{b}}_{q(j+2)}, \hat{\mathbf{b}}_{q(j+3)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+3)}) + d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+4)}) + d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+4)}) + d(\hat{\mathbf{b}}_{q(j+3)}, \hat{\mathbf{b}}_{q(j+4)}) \quad (4)$$

In determining $\Sigma(j\rightarrow3)$ and $\Sigma(j\rightarrow4)$, non-consecutive distances are considered, for instance $d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+4)})$ in $\Sigma(j\rightarrow4)$ to assess if the vectors are moving away from or back to the first multivariate positions.

Stage 5 to 9

The same data processing stages were performed when 4 or 5 samplings were considered being exchanged, $\Sigma(j\rightarrow2)$ for $\Sigma(j\rightarrow3)$ or $\Sigma(j\rightarrow4)$, and the respective maximum values and thresholds, i.e., $\Sigma_{Max}(j\rightarrow3)$ and $\Sigma_{TH}(j\rightarrow3)$ or $\Sigma_{Max}(j\rightarrow4)$ and $\Sigma_{TH}(j\rightarrow4)$, were determined. The thresholds are larger when more samplings and Euclidean distances are considered.

Stage 10

To assess hematological perturbations from 4 or 5 consecutive samplings, the same identification performance parameters were determined (i.e., TP, FP and LR).

Multivariate trends normalized for days difference

In the previously described determination of $\Sigma(j\rightarrow m-1)$, the days difference between samplings were not considered. Therefore, it is not possible to distinguish cases where the same multivariate change occurred in a few days or various months.

Thus, the assessment of multivariate trends normalized for days differences was also studied. Eq.▶ **(5)** describes the determination of a Euclidean distance normalized for days difference, $d^N(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})$, between vectors of normalized blood parameters ($\hat{\mathbf{b}}_{qj}$ and $\hat{\mathbf{b}}_{q(j+1)}$:

$$d^N(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) = \frac{d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})}{\Delta t\,(j, j+1)} \quad (5)$$

where $\Delta t(j, j+1)$ is the days difference between both sampling occasions.

The assessment of multivariate trends normalized for days difference is equivalent to the one described in the sections on multivariate trends of three samplings and multivariate trends of four or 5 samplings, but Euclidean distances are exchanged for normalized Euclidean distances. The normalized Euclidean distances are smaller than non-normalized ones.

When normalized Euclidean distances are summed, a normalized sum $\Sigma^N(j\rightarrow m-1)$ is estimated. For instance, for m = 3, $\Sigma^N(j\rightarrow2)$ is estimated by Eq.▶ **(6)**.

$$\Sigma^N(j\rightarrow2) = \frac{d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)})}{\Delta t\,(j, j+1)} + \frac{d(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)})}{\Delta t\,(j+1, j+2)} + \frac{d(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)})}{\Delta t\,(j, j+2)} = d^N(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+1)}) + d^N(\hat{\mathbf{b}}_{q(j+1)}, \hat{\mathbf{b}}_{q(j+2)}) + d^N(\hat{\mathbf{b}}_{qj}, \hat{\mathbf{b}}_{q(j+2)}) \quad (6)$$

The maximum value of various $\Sigma^N(j\rightarrow2)$, i.e., $\Sigma^N_{Max}(j\rightarrow2)$, represents the normalized hematological perturbation index (HP^N_{IX}).

Blood collection and analysis

Blood collections were performed as defined by WADA [13]. The blood samples were preserved at 4 °C and analyzed within 24 h after collection. The blood analyses were performed in an automatic hematology analyzer (Sysmex XT-2000i hematology analyzer; Sysmex Corp., Kobe, Japan) subject to regular internal and external quality controls. ▶ **Table 2** presents the relative reproducibility standard deviation (s'_R) of determinations estimated from various proficiency tests organized by the Quality Control Center Switzerland in 2018. These s'_R are supralaboratorial determinations of measurement standard uncertainty [22].

Athlete population and data confidentiality

The tool developed to determine hematological perturbation indexes was applied to data from 127 healthy athletes of different sports, genders, and ethnic groups for whom there was no prior information regarding doping. From this list of individuals, seven were prosecuted for doping based on their biological passport, i.e., based on the interpretation of hematological data by various expert hematologists. It was not possible to confirm the type of doping of the seven prosecuted athletes. A total of 41 of the studied athletes presented biological passports with negligible hematological perturbations assessed by a hematologist expert. The seven prosecuted athletes were considered positive controls and used to estimate the perturbation index thresolds (i.e., $\Sigma_{TH}(j\rightarrow m-1)$ and

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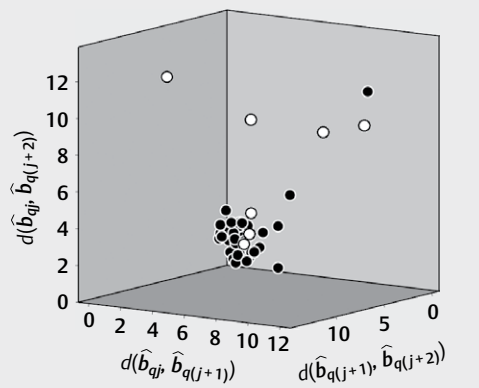
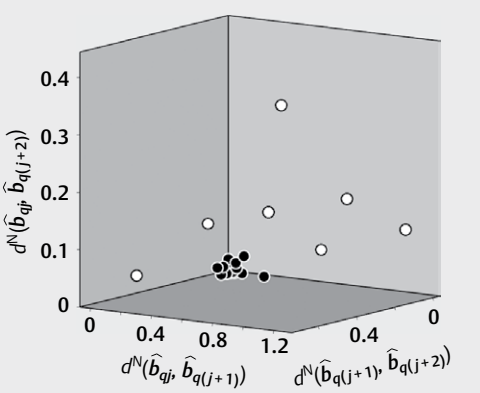


▶ **Fig. 1** Representation of Euclidian distances not normalized for days differences used to estimate $\Sigma_{Max}(j\rightarrow 2)$ from three consecutive sampling occasions. Cases with relevant or negligible perturbations are distinguished by point colors.



▶ **Fig. 2** Representation of Euclidian distances normalized for days differences used to estimate $\Sigma^{N}_{Max}(j\rightarrow 2)$ from three consecutive sampling occasions. Cases with relevant or negligible perturbations are distinguished by point colors.

▶ **Table 3** Correlation matrix of Euclidian distances not normalized for days differences determined from three consecutive samplings and eight blood parameters. The correlation coefficients are significantly different from 0 (p-value < 0.01).

	$d(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$
$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	0.548	
$d(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.712	0.582

$\Sigma^{N}_{TH}(j\rightarrow m-1)$ and the TP of identifications. The 41 athletes with negligible hematological perturbations were negative controls used to estimate the FP of perturbation identifications.

▶ **Table 4** Correlation matrix of Euclidian distances normalized for days differences determined from three consecutive samplings and 8 blood parameters. The correlation coefficients are significantly different from 0 (p-value < 0.01).

	$d^{N}(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	$d^{N}(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$
$d^{N}(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	0.525	
$d^{N}(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.833	0.776

The athlete's biological passports were processed after codes were randomly attributed to each individual, where the first seven codes were attributed to athletes prosecuted for doping.

This research followed the ethical standards of the International Journal for Sports Medicine [23].

Spreadsheet

The supplementary material includes the validated MS Excel files used in the research. The file does not include the data from the athletes studied.

Results and Discussion

Prior to the determination of hematological perturbation indexes, a qualitative assessment was performed of multivariate differences of biological passports of the 7 and 41 athletes presenting relevant and negligible perturbations of their organism, respectively. This assessment was performed by the graphical representation of multivariate trends collected to estimate $\Sigma_{Max}(j\rightarrow 2)$, $\Sigma^{N}_{Max}(j\rightarrow 2)$, $\Sigma_{Max}(j\rightarrow 4)$ and $\Sigma^{N}_{Max}(j\rightarrow 4)$ from the 8 blood parameters studied.

Qualitative assessment of multivariate hematological perturbations

Multivariate trends of three samplings

Because the determination of $\Sigma_{Max}(j\rightarrow 2)$ and $\Sigma^{N}_{Max}(j\rightarrow 2)$ involves calculating only three Euclidean distances, a 3D graph can be used to represent the data required to calculate these hematological perturbation indexes. ▶ **Fig. 1** represents the graph $d(\hat{b}_{qj}, \hat{b}_{q(j+1)})$ vs. $d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$ vs. $d(\hat{b}_{qj}, \hat{b}_{q(j+2)})$ used in the determination of each $\Sigma_{Max}(j\rightarrow 2)$. ▶ **Fig. 2** represents the graph $d^{N}(\hat{b}_{qj}, \hat{b}_{q(j+1)})$ vs. $d^{N}(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$ vs. $d^{N}(\hat{b}_{qj}, \hat{b}_{q(j+2)})$ used to calculate the various $\Sigma^{N}_{Max}(j\rightarrow 2)$. ▶ **Fig. 1** and ▶ **2** show that there are multivariate differences between the two groups of athletes, i.e., athletes in whom relevant and negligible hematological perturbations were confirmed by hematologists. However, the discrimination of these groups improves when Euclidian distances are normalized for the days difference between blood collections. In ▶ **Fig. 2**, the data from athletes with negligible hematological perturbations form an isolated cluster with smaller normalized Euclidian distances. The correlation matrices between variables represented in ▶ **Figs. 1** and **2** (▶ **Tables 3** and **4**) show significant correlation between variables with p-values smaller than 0.01. Pearson's correlation coefficients vary between 0.525 and 0.833.

Multivariate trends of five samplings

When multivariate trends of five consecutive samplings were studied by calculating the eight Euclidean distances required to determine $\Sigma_{Max}(j\rightarrow 4)$ or $\Sigma^{N}_{Max}(j\rightarrow 4)$, the qualitative assessment of multi-

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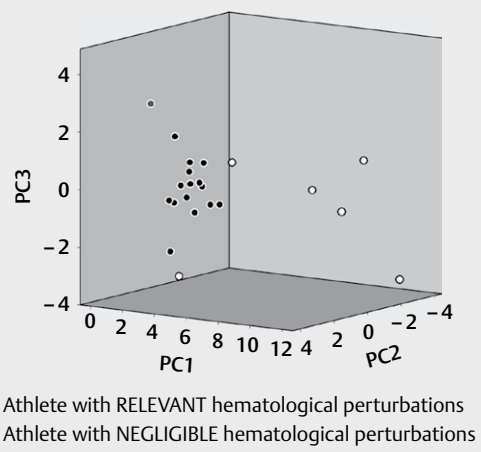
▶ **Table 5** Correlation matrix of Euclidian distances not normalized for days differences determined from 5 consecutive samplings and 8 blood parameters. The correlation coefficients not significantly different from 0 (i.e., p-value > 0.01) are identified with the p-value.

	$d(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	$d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	$d(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	$d(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	$d(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$
$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	0.628						
$d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	0.215 (p = 0.17)	0.445 (p = 0.02)					
$d(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.914	0.559	0.249 (p = 0.13)				
$d(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	0.926	0.637	0.368 (p = 0.05)	0.931			
$d(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	0.872	0.606	0.308 (p = 0.08)	0.926	0.927		
$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$	0.757	0.592	0.331 (p = 0.07)	0.751	0.846	0.784	
$d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$	0.397 (p = 0.03)	0.179 (p = 0.21)	0.344 (p = 0.06)	0.329 (p = 0.07)	0.459 (p = 0.02)	0.192 (p = 0.20)	0.402 (p = 0.03)



▶ **Fig. 3** Scores of the first three principal components (PC1, PC2 and PC3) of $\Sigma_{Max}(j \rightarrow 4)$ determined from five sampling occasions and eight blood parameters. The $\Sigma_{Max}(j \rightarrow 4)$ are not normalized for days difference between samplings.

▶ **Table 6** Score coefficients used to calculate scores of the first three principal components (PC1, PC2 and PC3) of Euclidian distances not normalized for days differences determined from five consecutive samplings and eight blood parameters.

	PC1	PC2	PC3
$d(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	0.178	−0.172	−0.152
$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	0.139	0.088	0.571
$d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	0.083	0.684	0.509
$d(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.177	−0.206	−0.104
$d(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	0.187	−0.049	−0.105
$d(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	0.177	−0.232	0.113
$d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$	0.167	−0.008	−0.066
$d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$	0.087	0.568	−0.750

variate trends was performed by examining the first three components of a principal component analysis. The summary of information provided in the first three principal components allows the graphical representation of the difference between the two groups of athletes.

When the 8 Euclidian distances are not normalized for days differences and the 8 blood parameters are considered, the first three principal components described 89 % of the total variance of the original data.

▶ **Table 5** presents the correlation matrix of the eight non-normalized Euclidian distances where a relevant direct correlation was observed between the various pairs of distances except for $d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$ and $d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$. When any Euclidian distance increases, all the others tend to increase where the correlation with $d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$ and $d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$ is not as strong. Those distances represent the distance between the 3rd and 4th and 4th and 5th sampling occasions. This observation suggests the distances between the first three sampling occasions contribute more to $\Sigma_{Max}(j \rightarrow 4)$.

▶ **Fig. 3** presents the graphical representation of the scores of $\Sigma_{Max}(j \rightarrow 4)$ for the first three components (PC1, PC2 and PC3) of a principal components analysis. ▶ **Table 6** presents the calculated score coefficients. ▶ **Fig. 3** shows that PC1 can be used to distinguish athletes with negligible and relevant hematological perturbations. When PC1 increases, the hematological system perturbation is larger. The positive score coefficients of PC1 (▶ **Table 6**) prove that PC1 is larger when all Euclidian distances are larger ($PC1 = 0.178 \cdot d(\hat{b}_{qj}, \hat{b}_{q(j+1)}) + 0.139 \cdot d(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)}) + (\ldots) + 0.087 \cdot d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$. All Euclidian distances are equally relevant for PC1 except $d(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$ and $d(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$, which are associated with half-score coefficients. Therefore, it can be concluded that the Euclidian distances not normalized for days difference can distinguish athletes affected by negligible or relevant hematological perturbations.

When the 8 Euclidian distances normalized for the days differences and based on 8 blood parameters are considered, the first three principal components represent 92 % of data variance. ▶ **Table 7** presents the correlation matrix of Euclidian distances normalized for days difference between samplings. ▶ **Table 8** presents the score coefficients of these principal components (PC1, PC2 and PC3).

▶ **Table 7** shows a relevant direct correlation between most Euclidian distances where no specific distance seems to be independent of the others, such as for two distances not normalized for days difference. Therefore, a comparison of ▶ **Table 5** with ▶ **Table 7** suggests days normalization fixed some lacks of correlation of non-normalized distances.

▶ **Fig. 4** presents the scores of PC1, PC2 and PC3 for $\Sigma^{N}_{Max}(j \rightarrow 4)$ estimated from score coefficients of ▶ **Table 8**. It can be observed from this figure that when athletes have their hematological system disturbed, PC1 is larger, and PC2 and PC3 are away from zero, either above or below zero. The data normalized for days difference

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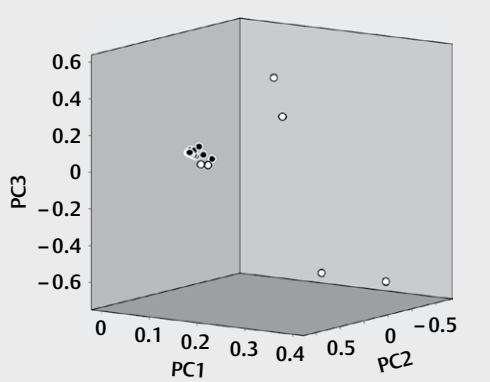
▶ **Table 7** Correlation matrix of Euclidian distances normalized for days differences determined from five consecutive samplings and eight blood parameters. The correlation coefficients not significantly different from 0 (i.e. with p-value > 0.01) are identified with the respective p-value.

	$d^N(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	$d^N(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	$d^N(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	$d^N(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	$d^N(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$
$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	−0.019 (p = 0.47)						
$d^N(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	−0.046 (p = 0.42)	0.228 (p = 0.15)					
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.639	0.530	0.404 (p = 0.03)				
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	0.420 (p = 0.03)	0.497	0.645	0.908			
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	0.733	0.329 (p = 0.07)	0.476 (p = 0.01)	0.941	0.889		
$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$	0.205 (p = 0.18)	0.539	0.545	0.797	0.928	0.725	
$d^N(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$	0.009 (p = 0.48)	0.690	0.721	0.574	0.742	0.526	0.630

▶ **Table 8** Score coefficients used to calculate scores of the first three principal components (PC1, PC2 and PC3) of Euclidian distances normalized for days differences determined from five consecutive samplings and 8 blood parameters.

	PC1	PC2	PC3
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+1)})$	0.089	0.543	0.037
$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+2)})$	0.120	−0.262	0.811
$d^N(\hat{b}_{q(j+2)}, \hat{b}_{q(j+3)})$	0.128	−0.274	−0.724
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+2)})$	0.185	0.182	0.164
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+3)})$	0.195	0.010	−0.113
$d^N(\hat{b}_{qj}, \hat{b}_{q(j+4)})$	0.179	0.251	−0.118
$d^N(\hat{b}_{q(j+1)}, \hat{b}_{q(j+4)})$	0.176	−0.075	0.020
$d^N(\hat{b}_{q(j+3)}, \hat{b}_{q(j+4)})$	0.156	−0.316	0.011



▶ **Fig. 4** Scores of the first three principal components (PC1, PC2 and PC3) of $\Sigma^N_{Max}(j\rightarrow4)$ determined from five sampling occasions and eight blood parameters. The $\Sigma^N_{Max}(j\rightarrow4)$ are normalized for days difference between samplings.

allows isolating, in a small cluster, the scores of athletes with negligible hematological perturbations. The score coefficients from ▶ **Table 8** support the findings of ▶ **Fig. 4** because all PC1 coefficients are positive and some PC2 and PC3 coefficients are positive or negative.

The comparison of the multivariate data used to estimate $\Sigma_{Max}(j\rightarrow4)$ and $\Sigma^N_{Max}(j\rightarrow4)$ suggests the Euclidian distances normalized for days differences improves the ability to distinguish negligible and relevant hematological perturbations.

Quantitative analysis of multivariate hematological perturbations

After the interpretation of ▶ **Figs. 1–4**, it becomes clear that the collected Euclidian distances can express useful multivariate information to distinguish relevant perturbations of the hematological system of athletes. The normalization of that information for sampling days differences improves the ability to distinguish hematological perturbations.

▶ **Tables 9–11** present index thresholds, TP, FP, and LR from identifications of relevant hematological perturbations in athletes when 3, 4, or 5 sampling occasions, respectively, are considered. Each table shows the performance from Euclidian distances non-normalized or normalized for days differences between sampling occasions and considers vectors from only two or more blood parameters. The selected parameters are divided into Case 1–6.

The identification performance was determined from 300 simulations of equivalent hematological histories of each athlete given the uncertainty of blood parameter measurements. Because the reproducibility standard deviation overestimates the measurement uncertainty [22], the FPs are overestimated, i.e., larger FPs were estimated than would be from not overestimated measurement uncertainty. When no simulated false positive result is observed, the FP is overestimated by the ratio between 1 and the total number of simulations (i.e., 300 times the number of studied athletes with negligible hematological perturbations). The FPs from identifications based on 3, 4, or 5 sampling occasions were estimated from data of 41, 24, and 16 athletes, respectively. The simulation allows extracting more information from the available data.

The index threshold (Σ_{TH}) was set to allow TPs larger than 98 %, and subsequently the FP and LR (LR = TP/FP) were determined. Typically, the definition of the threshold involved several attempts.

The performance of the identifications of relevant hematological perturbations was compared by taking the LR into account because thresholds are associated with a different TP.

The Σ_{TH} is observed to increase as the number of considered blood parameters also increases, and Σ_{TH} is larger for data not normalized for days difference, as expected.

The identification performance improves (i.e., the LR increases) when Euclidian distances are normalized for days differences where this improvement is more obvious for four and five sampling occasions. For more sampling occasions, the identification performance

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▶ **Table 9** Thresholds and performance of the identification of doping from the determination of hematological perturbation indexes from three sampling occasions. Data from 48 athletes was considered.

	Selected parameters					
Parameter	**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5**	**Case 6**
HGB	X	X	X	X	X	X
HCT %		X	X	X	X	X
RBC		X	X	X	X	X
RET						X
RET%	X	X	X	X	X	X
MCV			X	X	X	X
MCH				X	X	X
MCHC					X	X
Non-normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 2)$	4.5	7.2	7.5	7.5	8.1	8.8
TP (%)	99.10	98.76	98.71	99.48	99.38	99.48
FP (%)	1.14	0.07	0.02	0.54	1.03	0.66
LR	87	1518	6071	185	96	151
Normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 2)$	0.13	0.23	0.23	0.23	0.23	0.23
TP (%)	99.95	99.67	99.57	99.67	99.86	99.67
FP (%)	0.21	0.01	0.03	0.14	0.73	0.90
LR	473	12259	3062	721	136	110

▶ **Table 10** Thresholds and performance of the identification of doping from the determination of hematological perturbation indexes from four sampling occasions. Data from 31 athletes was considered.

	Selected parameters					
Parameter	**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5**	**Case 6**
HGB	X	X	X	X	X	X
HCT %		X	X	X	X	X
RBC		X	X	X	X	X
RET						X
RET%	X	X	X	X	X	X
MCV			X	X	X	X
MCH				X	X	X
MCHC					X	X
Non-normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 3)$	6.5	9.0	9.6	10	11	11.5
TP (%)	99.76	99.57	99.05	99.05	99.33	99.86
FP (%)	2.33	2.75	2.47	3.00	5.90	7.79
LR	43	36	40	33	17	13
Normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 3)$	0.15	0.25	0.25	0.26	0.26	0.26
TP (%)	99.00	99.57	99.68	98.76	99.33	99.90
FP (%)	0.01	0.01	0.01	0.01	0.01	0.10
LR	7128	7169	7177	7111	7152	1028

does not change significantly when more blood parameters are considered.

For normalized hematological perturbation indexes, the identifications from four samplings are better than identifications based on 5 samplings, except for Case 6 where all blood parameters are considered. The LRs from 4 and 5 samplings were compared via a t-test for a 99 % confidence level using the standard deviation of estimated LR. This difference is attributed to the rather low and different number of athletes who have data of four or five samplings available (i.e., 24 and 16, respectively).

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▶ **Table 11** Thresholds and performance of the identification of doping from the determination of hematological perturbation indexes from five sampling occasions. Data from 22 athletes was considered.

	Selected parameters					
Parameter	**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5**	**Case 6**
HGB	X	X	X	X	X	X
HCT %		X	X	X	X	X
RBC		X	X	X	X	X
RET						X
RET%	X	X	X	X	X	X
MCV			X	X	X	X
MCH				X	X	X
MCHC					X	X
Non-normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 4)$	10.5	15.5	15.5	16.5	18	19
TP (%)	99.71	98.90	99.71	99.10	98.95	99.57
FP (%)	1.08	0.02	0.10	0.29	1.00	1.56
LR	92	4747	957	340	99	64
Normalized index						
$\Sigma^{N}_{TH}(j\rightarrow 4)$	0.35	0.45	0.5	0.5	0.6	0.84
TP (%)	98.52	98.67	98.95	99.81	99.90	97.76
FP (%)	0.02	0.02	0.02	0.02	0.02	0.02
LR	4729	4736	4750	4791	4795	4693

In the future, the tool developed will be evaluated with more data until thresholds and performance characteristics become stable, i.e., independent of available data.

However, at this stage, it is possible to know that the developed algorithms and defined thresholds can be used as an early warning system of suspected doping with acceptable rates of false responses. The only consequence of a false positive from this tool is the waste of the hematologist's time in analyzing negligible hematological perturbations. However, as more data is fed into the model, better control of false responses will be achieved.

Conclusions

The strategy developed for quantifying multivariate variations of blood parameters in samples of athletes collected on more than two occasions was successfully applied to distinguish two groups of athletes experiencing relevant or negligible hematological perturbations. This tool can be used for early identification of cases that should be more closely monitored or forwarded to expert hematologists to decide on prosecution for doping. The assessment of the multivariate trends can also be used to detect health problems that impact the hematological system.

Hematological perturbation monitoring involves quantifying Euclidian distances between vectors of various blood parameters normalized for sampling days difference. Several normalized Euclidian differences from m sampling occasions (i.e., blood parameter vectors) are summed and the maximum sum, calculated from more than m samplings, is defined as the hematological perturbation index of the athlete. A threshold for this index was estimated by simulating equivalent hematological histories of the seven available cases prosecuted for doping. Monte Carlo simulations of the blood parameters were performed by taking into account the uncertainty of blood parameter determination. This uncertainty was overestimated by the reproducibility standard deviation observed in international proficiency tests. A threshold for a true positive result rate (TP) larger than 98 % was defined.

The simulation of data from 41 to 16 athletes with negligible hematological perturbations, given the uncertainty of the blood parameter determination, was used to estimate the false positive result rates (FPs). The FP represents the chance of these hematological profiles producing a hematological perturbation index larger than the defined threshold.

The tools developed were applied to the identification of hematological perturbations from three, four, or five sampling occasions. The low FP of identifications (between 0.9 % and 0.01 %) proves this tool can be used for early identification of suspected doping.

More data from athletes with relevant and negligible hematological perturbations should be used to make defined thresholds and estimated performance parameters applicable to a wider diversity of cases.

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Conflict of Interest

Authors declare that they have no conflict of interest.

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